
Trust Me or Check Me? Self-Reported Confidence, Verification, and Decision Authority in Language Models

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Abstract

1

2 1 Introduction

3 Large language models are increasingly asked not only to produce answers, but also to influence
4 whether those answers receive further scrutiny. A system may answer directly, seek additional
5 information, invoke a tool, defer to a human, or recommend that an answer be independently checked
6 before it is used. This creates a second-order decision problem beyond answer generation itself: *when*
7 *should an already-generated answer be used as is, and when should it be verified first?* When the
8 same AI system both generates an answer and participates in deciding whether that answer needs
9 checking, uncertainty becomes part of a decision-authority mechanism: the system can shape not
10 only what information is produced, but also when independent judgment enters the loop.

11 A natural control signal for this decision is the model’s own reported confidence. Suppose a model
12 has already produced an answer and reports a probability q that the answer is correct. If independent
13 verification costs C , while using a wrong answer without verification incurs cost L , then treating q as
14 the probability of correctness induces the simple expected-cost rule

$$(1 - q)L > C \implies \text{VERIFY_FIRST.}$$

15 This apparently straightforward rule hides two different questions. First, will a model’s downstream
16 verification behavior actually follow the policy implied by its stated confidence? Second, if it does,
17 will stronger agreement with that policy produce better decisions about which answers actually need
18 checking? These questions are distinct. A verification policy can be highly consistent with reported
19 confidence even when that confidence is poorly aligned with whether a particular answer is correct.

20 We study these questions with a controlled post-answer design that separates answer generation,
21 confidence elicitation, and verification policy. For each question, the model first produces a multiple-
22 choice answer, which is frozen. It then reports a probability q that this frozen answer is correct, which
23 is also frozen. Only afterward does the model choose between using the answer without verification
24 and verifying it first. At this final stage, we vary whether the previously reported q is hidden or
25 explicitly shown, whether verification authority is described as belonging to a human or to the AI
26 system, and the cost of using a wrong answer without verification. Because the answer and reported
27 confidence are held fixed before these manipulations, the experiment isolates changes in downstream
28 verification policy rather than changes in answer generation or confidence elicitation.

29 We evaluate 500 deterministic MMLU-Pro questions across four model families: OpenAI GPT-5.6
30 Sol, Anthropic Claude Sonnet 5, Google Gemini 3.8 Flash, and xAI Grok 4.20 non-reasoning.
31 For each model-question pair, the final decision crosses two decision-authority conditions, two
32 confidence-visibility conditions, and four error costs $L \in \{2, 5, 10, 20\}$, with verification cost fixed

33 at $C = 1$. This yields 32,000 primary verification decisions. Statistical comparisons use the question
34 as the independent sampling unit and question-level paired bootstrap confidence intervals.

35 Explicitly surfacing the model’s own previously reported confidence strongly changes verification
36 behavior, but not in a common direction across model families. For Claude Sonnet 5, visible
37 confidence increases verification by approximately 12.8–22.4 percentage points across authority and
38 cost conditions. For GPT-5.6 Sol, it decreases verification by 16.2–20.8 points throughout. Grok
39 4.20 similarly decreases verification by 8.6–16.4 points. Gemini 3.8 Flash shows smaller effects at
40 low error costs but substantially higher verification when confidence is visible at larger costs, with
41 increases of roughly 12–17 points at $L = 10$ and $L = 20$. Thus, the same intervention— making
42 self-reported confidence explicit to the downstream decision—can produce large and opposite policy
43 shifts across model families.

44 The most revealing case is GPT-5.6 Sol. We define *confidence-policy alignment* by comparing the
45 model’s final action with the action implied by its frozen reported confidence under the supplied cost
46 structure. When confidence is hidden, GPT-5.6 Sol disagrees with this confidence-implied policy
47 on roughly 19.6–27.2% of questions across authority and cost conditions. When the same reported
48 q is explicitly shown, disagreement collapses to 0.2–2.2%. Its behavior simultaneously becomes
49 almost perfectly ordered by the cost of error: stake-monotonicity violations fall from 95 to 1 per
50 1,000 trajectories.

51 Yet this increased alignment with reported confidence does not improve actual error filtering. At
52 the highest error cost, among GPT-5.6 Sol answers that are in fact wrong, the fraction used without
53 verification rises from 32.18% to 56.32% under human decision authority and from 33.33% to
54 54.02% under AI decision authority when confidence is made visible. In this case, explicitly surfacing
55 confidence therefore produces a policy that is substantially more consistent with the model’s stated
56 uncertainty while leaving substantially more of its high-cost errors unchecked. Greater confidence-
57 policy alignment and better verification quality are not the same property.

58 This distinction also matters when comparing models. A system can catch nearly all of its wrong
59 answers simply by verifying almost everything, but such a policy defeats the purpose of selective
60 verification. Conversely, reducing verification burden can leave more errors unchecked. We therefore
61 evaluate verification behavior using both error filtering and unnecessary verification of correct answers
62 rather than interpreting raw verification rate as a measure of quality.

63 Decision authority produces a smaller and less universal effect. Claude shows the clearest directional
64 shift, generally verifying more when the AI system rather than the human is described as controlling
65 the decision, particularly at higher error costs. The other model families show small or mixed average
66 effects. We therefore do not find a general human-versus-AI authority asymmetry across models.
67 Finally, on a prespecified 100-question robustness subset, the major confidence-visibility directions
68 persist under one semantically equivalent Stage-3 paraphrase, although we do not claim invariance to
69 arbitrary prompt wording.

70 These results concern model behavior rather than human responses. We do not measure whether
71 people trust, follow, resist, or override a model’s verification recommendation. Instead, we isolate
72 a technical prerequisite for such questions: whether a model’s own uncertainty can reliably govern
73 when its outputs are subjected to independent scrutiny. This matters for human–AI systems in which
74 an AI does more than provide information and also participates in determining when human judgment,
75 external evidence, or additional computation should intervene.

76 **Contributions.** Our contributions are threefold. First, we introduce a controlled post-answer
77 evaluation that freezes both answer and reported confidence before separately varying confidence
78 visibility, decision authority, and error cost across 32,000 matched verification decisions. Second,
79 we show that explicitly surfacing self-reported confidence can strongly reshape verification policy
80 in opposite directions across model families, cautioning against treating reported confidence as
81 a universally portable verification-control signal. Third, we identify a direct divergence between
82 confidence-policy alignment and verification quality: for GPT-5.6 Sol, visible confidence makes
83 downstream actions nearly coincide with the policy implied by reported q while substantially more
84 wrong high-cost answers remain unverified.

85 2 Related Work

86 **Self-reported uncertainty in language models.** A growing literature studies whether language
87 models can express useful uncertainty about their own answers. Early work showed that models
88 can be trained to verbalize probabilities that are meaningfully calibrated [Lin et al., 2022], while
89 subsequent studies examined black-box confidence elicitation in instruction-tuned and closed-source
90 models [Tian et al., 2023, Xiong et al., 2024]. These results motivate self-reported confidence
91 as an accessible uncertainty signal, particularly when token probabilities or model internals are
92 unavailable. At the same time, calibration is not sufficient for our setting: even a confidence score
93 that is informative about correctness may or may not govern a model’s later decision about whether
94 its answer should be checked.

95 **Confidence, abstention, and downstream action.** Recent work has begun to examine this gap
96 directly. RiskEval evaluates whether LLMs adapt abstention behavior as the penalty for an incorrect
97 answer changes, finding that models can fail to convert verbal uncertainty into appropriately risk-
98 sensitive abstention policies [Wang et al., 2026]. Kumaran [Kumaran, 2026] uses a two-stage answer-
99 confidence-abstention paradigm and finds that verbal confidence predicts subsequent commitment
100 more strongly than objective correctness, distinguishing verbal confidence from token-probability-
101 based uncertainty. More broadly, selective prediction and abstention methods use uncertainty to
102 trade coverage for error reduction, including approaches that explicitly control the error rate among
103 accepted answers [Dong and Shinnou, 2026].

104 Our experiment is complementary to this work. Rather than asking only whether reported confidence
105 predicts a later decision, we freeze the answer and reported probability and intervene on whether
106 that same probability is explicitly available at decision time. This permits a matched comparison of
107 the downstream policy with confidence hidden versus visible. It also separates *confidence-policy*
108 *alignment*—whether the action agrees with the decision rule implied by reported q —from *verification*
109 *quality*—whether wrong answers are filtered while unnecessary verification of correct answers is
110 avoided.

111 **Reliance and decision authority in human–AI systems.** Research on AI-assisted decision making
112 distinguishes mere reliance from *appropriate* reliance: accepting useful AI advice while rejecting
113 or scrutinizing erroneous advice [Schemmer et al., 2022]. Interface interventions can alter this
114 behavior; for example, cognitive forcing mechanisms can reduce human overreliance on incorrect AI
115 recommendations [Bućinca et al., 2021]. Our study does not measure human reliance. Instead, it
116 examines an upstream system-design question relevant to such interactions: when an AI participates
117 in deciding whether its own output deserves independent scrutiny, how do its reported uncertainty
118 and the location of decision authority shape that recommendation?

119 3 Experimental Setup

120 3.1 Dataset and models

121 We use the MMLU-Pro test split [Wang et al., 2024], a multiple-choice benchmark spanning 14
122 subject categories. The primary V2-B sample contains 500 questions: the earlier 200-question V2-A
123 sample plus 300 additional questions selected by deterministic category-stratified sampling without
124 reference to observed model results. The 500 questions constitute a single primary sample; V2-A and
125 V2-B are therefore not independent replications. A separate 100-question category-stratified subset,
126 also selected without reference to outcomes, is used for prompt robustness.

127 We evaluate four independently developed model families: OpenAI GPT-5.6 Sol (gpt-5.6-sol),
128 Anthropic Claude Sonnet 5 (claude-sonnet-5), Google Gemini 3.8 Flash (gemini-3.8-flash),
129 and xAI Grok 4.20 non-reasoning (grok-4.20-0309-non-reasoning). OpenAI used reasoning
130 effort none, Anthropic thinking was disabled, Gemini used thinking level low, and the xAI model
131 is explicitly non-reasoning. No tools, search, retrieval, code execution, or external context were
132 available. We used provider-native structured JSON outputs and did not override temperature or top- p
133 sampling parameters. Output was capped at 64 tokens for OpenAI, Anthropic, and xAI. Gemini used
134 a 512-token ceiling because its mandatory low-thinking tokens share the output budget.

3.2 Three-stage protocol

Each model–question pair proceeds through three stages. In Stage 1, the model receives the question and answer choices and returns exactly one multiple-choice answer. This answer is then frozen. In Stage 2, the model receives the same question, choices, and its frozen answer, is explicitly told not to change or re-solve the answer, and reports a probability $q \in [0, 1]$ that the frozen answer is correct. This reported confidence is also frozen.

Stage 3 does not permit the answer to change. Instead, the model selects exactly one of two actions: `USE_UNVERIFIED` or `VERIFY_FIRST`. Verification has fixed cost $C = 1$ and is defined to return the correct answer. Using the frozen answer without verification costs zero if it is correct and L if it is wrong, where

$$L \in \{2, 5, 10, 20\}.$$

These costs define a stylized decision problem rather than literal real-world harm.

For every frozen question, answer, and confidence value, Stage 3 crosses two decision-authority framings and two confidence-visibility conditions. Under *human authority*, the prompt states that the human controls whether the answer is independently verified before use; under *AI authority*, it states that the AI system controls verification before delivery to the human. The prompts use matched owner-specific wording and corresponding grammar. In the *visible-confidence* condition, Stage 3 additionally states the model’s previously reported probability q ; in the *hidden-confidence* condition, this sentence is omitted. Confidence is never re-elicited in Stage 3. Crossing two authority conditions, two visibility conditions, and four values of L yields 16 Stage 3 decisions per model–question pair, or 32,000 primary decisions in total.

3.3 Metrics

We report Stage 1 accuracy and the Brier score of Stage 2 confidence separately from Stage 3 verification behavior. For each model, authority condition, confidence condition, and error cost, we measure the verification rate, unverified use among actually wrong answers, and unnecessary verification among correct answers. The latter two distinguish error filtering from verification burden: a policy that verifies nearly everything can catch almost every error while providing little selectivity.

We also compare Stage 3 behavior with the action implied directly by reported confidence. Under the supplied cost model, the confidence-implied policy verifies when

$$(1 - q)L > C.$$

We call agreement between the model’s Stage 3 action and this action *confidence-policy alignment*, and report the corresponding disagreement rate. This is a behavioral comparison with the model’s *reported* probability; it does not assume that q represents a latent internal belief. We additionally measure stake-monotonicity violations, defined as `VERIFY_FIRST`→`USE_UNVERIFIED` reversals as L increases within an otherwise matched trajectory, as well as realized cost under the synthetic cost model.

The confidence-visibility effect is the paired difference

$$P(\text{VERIFY_FIRST} \mid \text{visible}) - P(\text{VERIFY_FIRST} \mid \text{hidden}),$$

and the owner effect is

$$P(\text{VERIFY_FIRST} \mid \text{AI authority}) - P(\text{VERIFY_FIRST} \mid \text{human authority}).$$

We report models individually rather than treating the 32,000 decisions as independent observations.

3.4 Statistical inference and prompt robustness

The independent sampling unit is the question. We use 5,000 question-level paired bootstrap resamples and report nominal 95% percentile confidence intervals, keeping all matched factor cells for a resampled question together. For metrics restricted to wrong answers, questions are resampled first and the wrong-answer subset is applied within each bootstrap replicate. We do not apply a multiplicity correction across the many descriptive cellwise comparisons; accordingly, we emphasize effect sizes, intervals, and repeated directional patterns rather than isolated significant cells.

Table 1: Stage 1 answer accuracy and Stage 2 confidence calibration on the 500-question primary sample. Brier score is computed from each model’s reported probability that its frozen answer is correct.

Model	Accuracy (%)	Wrong n	Brier ↓
Gemini 3.8 Flash	88.0	60	0.0834
GPT-5.6 Sol	82.6	87	0.1579
Claude Sonnet 5	74.2	129	0.1633
Grok 4.20 non-reasoning	64.4	178	0.1912

Prompt robustness uses the fixed 100-question subset with the same frozen answers and reported confidences under one semantically equivalent Stage 3 paraphrase. The preregistered robustness analysis covered the hidden-confidence condition (3,200 decisions). After completion of the primary experiment, we ran the corresponding visible-confidence extension (3,200 additional decisions). We therefore distinguish the hidden-condition robustness analysis as preregistered and the visible-condition extension as post-primary.

4 Results

4.1 Baseline answer and confidence performance

The four model families differ substantially in both answer accuracy and reported-confidence calibration (Table 1). Gemini has the highest Stage 1 accuracy at 88.0%, followed by GPT-5.6 Sol at 82.6%, Claude Sonnet 5 at 74.2%, and Grok 4.20 at 64.4%. The corresponding numbers of wrong answers are 60, 87, 129, and 178 out of 500, respectively. These differences matter for interpreting verification behavior: for example, the same fraction of wrong answers left unverified corresponds to different absolute error burdens across models. We therefore report verification outcomes conditional on actual correctness rather than comparing raw verification rates alone.

4.2 Explicit confidence strongly changes verification policy

Making the model’s own frozen confidence explicit produces large changes in verification behavior, but the direction differs sharply across model families (Figure 1). Claude becomes consistently more verification-prone: across both authority conditions and all four error costs, visible confidence increases the verification rate by 12.8–22.4 percentage points. All eight corresponding 95% bootstrap intervals exclude zero.

GPT-5.6 Sol moves in the opposite direction. Showing its frozen confidence reduces verification in all eight matched authority–cost conditions, by 16.2–20.8 percentage points; again, all eight intervals exclude zero. Grok 4.20 shows the same qualitative direction, with decreases of 8.6–16.4 points across all eight conditions.

Gemini exhibits a different cost-dependent pattern. At $L = 2$, visible confidence slightly reduces verification by 2.0–2.2 points. At $L = 5$, the effects are small and mixed across authority conditions. At the larger error costs, however, visibility substantially increases verification: by 12.4–13.0 points at $L = 10$ and 16.4–16.6 points at $L = 20$, with all four high-cost intervals excluding zero.

The consistency of these within-model directions is more important than any single cellwise comparison. The same intervention—revealing a probability the model itself had already reported for the same frozen answer—increases verification by more than 20 points in some Claude conditions while decreasing it by roughly 20 points in GPT-5.6 Sol. Self-reported confidence therefore does not act as a model-independent verification-control signal with a common behavioral effect. Its downstream influence depends strongly on the model family using it.

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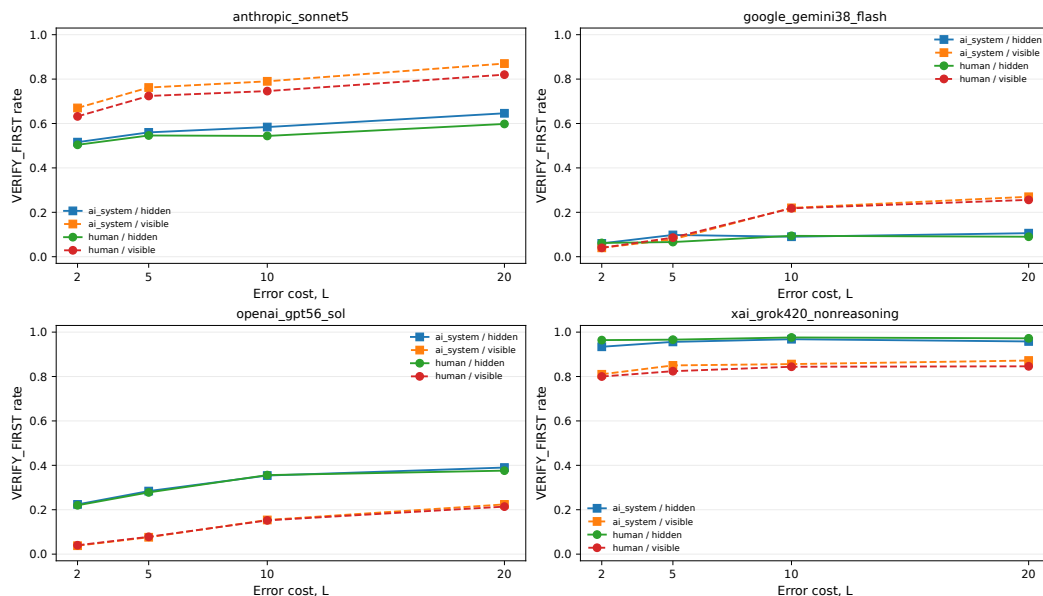


Figure 1: Verification rates across error costs, confidence visibility, and decision authority. Each panel shows one model family. Solid lines denote hidden confidence and dashed lines visible confidence; marker shape distinguishes human from AI decision authority. Explicitly surfacing the same previously reported confidence produces opposite shifts across model families: Claude generally verifies more, GPT-5.6 Sol and Grok 4.20 verify less, and Gemini shifts toward greater verification primarily at higher error costs.

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